Computer Science & Engineering Department

Arhitectura Sistemelor de Calcul

Curs 13
cs.pub.ro
cs.curs.pub.ro

University Politehnica of Bucharest



In cursul anterior

- Top 500 List
- Arhitecturi si Aplicatii Relevante



Cuprins

- Top 5
- Concluzii Top 500



Top 5

The HPCG Benchmark

- The High Performance Conjugate Gradients (HPCG) Benchmark
  - New metric for ranking HPC systems
  - **Complementary** to the High Performance LINPACK (HPL) benchmark
  - Designed to exercise computational and data access patterns that more closely match a different and broad set of important applications
  - Implementation written in C++ with MPI and OpenMP support
  - Offers extensive GPU support



The HPCG Benchmark

- HPCG is a stand-alone code measuring performance of basic operations:
  - Sparse matrix-vector multiplication
  - Vector updates
  - Global dot products
  - Local symmetric Gauss-Seidel smoother
  - Sparse triangular solve
  - Driven by multi-grid preconditioned conjugate gradient algorithm exercising key kernels on a nested set of coarse grids



Top 5 @HPCG

Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	HPCG (TFlop/s)
1	2	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	16004.50
2	1	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,194.00	14054.00
3	3	**LUMI** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,220,288	309.10	3408.47
4	4	**Leonardo** - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, Atos EuroHPC/CINECA Italy	1,824,768	238.70	3113.94
5	5	**Summit** - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	148.60	2925.75

JUNE 2023

			SITE	COUNTRY	CORES	RMAX PFLOP/S	POWER MW
1	**Frontier**	HPE Cray EX235a, AMD Opt 3rd Gen EPYC (64C 2GHz), AMD Instinct MI250X, Slingshot-11	DOE/SC/ORNL	USA	8,699,904	1,194.0	22.7
2	**Fugaku**	Fujitsu A64FX (48C, 2.2GHz), Tofu Interconnect D	RIKEN R-CCS	Japan	7,630,848	442.0	29.9
3	**LUMI**	HPE Cray EX235a, AMD Opt 3rd Gen EPYC (64C 2GHz), AMD Instinct MI250X, Slingshot-11	EuroHPC/CSC	Finland	2,220,288	309.0	6.01
4	**Leonardo**	Atos Bullsequana intelXeon (32C, 2.6 GHz), NVIDIA A100 quad-rail NVIDIA HDR100 Infiniband	EuroHPC/CINEC	Italy	1,824,768	238.7	7.40
5	**Summit**	IBM POWER9 (22C, 3.07GHz), NVIDIA Volta GV100 (80C), Dual-Rail Mellanox EDR Infiniband	DOE/SC/ORNL	USA	2,414,592	148.6	10.1

Top 5 @HPL

5 - Summit Power System AC922

- Site: DOE/SC/Oak Ridge National Laboratory
- Familia de sisteme: IBM Power Systems
- Procesor: IBM Power9 22C 3.1GHz, NVIDIA Volta V100
- OS: Red Hat Enterprise Linux
- Arhitectura: Cluster
- Aplicatii: Cercetare
- Anul instalarii: 2018
- Numar de core-uri: 2,414,592
- Rmax(PFlops): **148**,6
- Rpeak(PFlops): **200**,8
- HPCG(PFlops): **2,**93
- Memorie: **2,802** TB
- Consum: **10.096** kW
- Interconectare: Dual-rail Mellanox EDR Infiniband



Node Stats

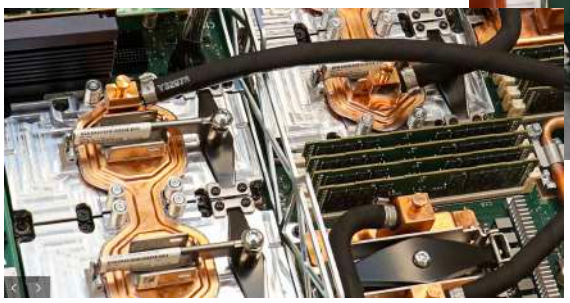
- Faster I/O - up to 5.6x more I/O bandwidth than x86 servers
- GPUs: 2-6 NVIDIA® Tesla® V100 GPUs with NVLink
- CPUs: 2x POWER9 CPUs
- Simplest AI architecture: Share RAM across CPUs & GPUs
- PCIe Gen4 2x faster vs PCIe Gen3 in x86

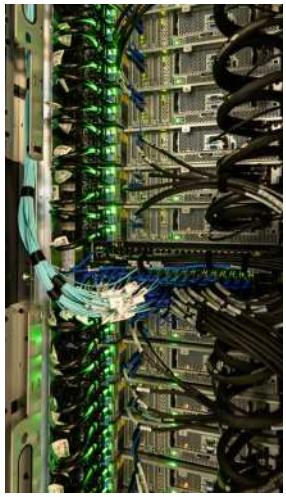


https://apps.kaonadn.net/4882011/IBMPower8/index.html#AC922;C1272

Summit Infrastructure



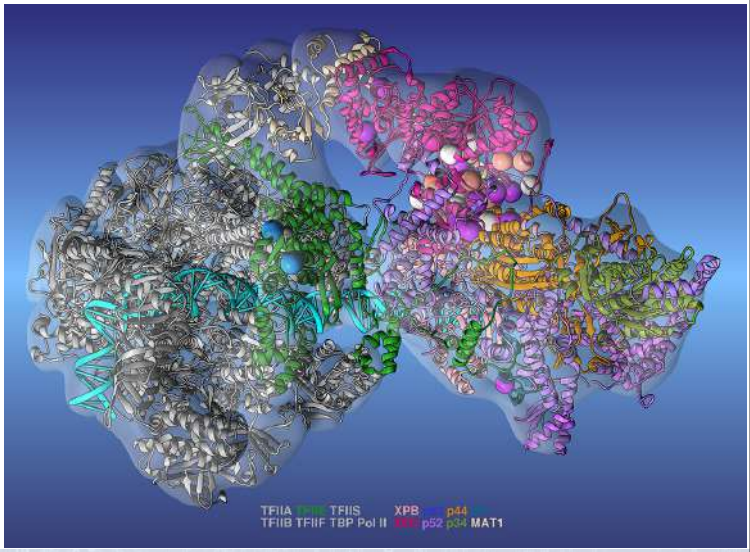




R&D Highlight

- Accurately identify clusters of gene mutations (spheres), which helps study the emergence of various genetic diseases
- From months to days using CryoEM & NAMD

https://www.olcf.ornl.gov/2019/05/20/summit-charts-a-course-to-uncover-the-origins-of-genetic-diseases/

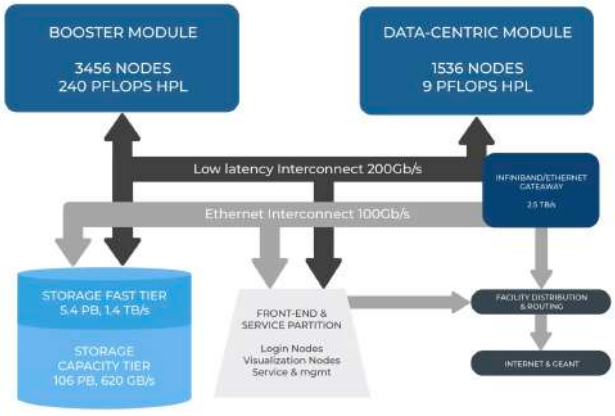


4 - Leonardo - BullSequana XH200

- Site: EuroHPC/CINECA
- Familia de sisteme: Atos
- Procesor: Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64GB
- OS: Linux
- Arhitectura: Cluster
- Aplicatii: Cercetare
- Anul instalarii: 2022
- Numar de core-uri: 1,824,768
- Rmax(PFlops): **238**,7
- Rpeak(PFlops): **304**,5
- HPCG(PFlops): **3,**11
- Memorie: **3,1** PB
- Consum: **7.404** kW
- Interconectare: Quad-Rail NVidia HDR 100 Infiniband



Leonardo Node Infrastructure



- 3456 booster nodes
  - Intel Xeon 8358 32 cores & 4XNvidia A100 64GB HBM2
- 1536 data-centric nodes
  - Intel Saphire Rapids 2x56 cores / 48x32 GB DDR5 4.8GHz
- 16 visualization nodes
  - 2 x Icelake 32cores 2.4GHz / 3 NVIDIA V100, RAM 16x32 GB

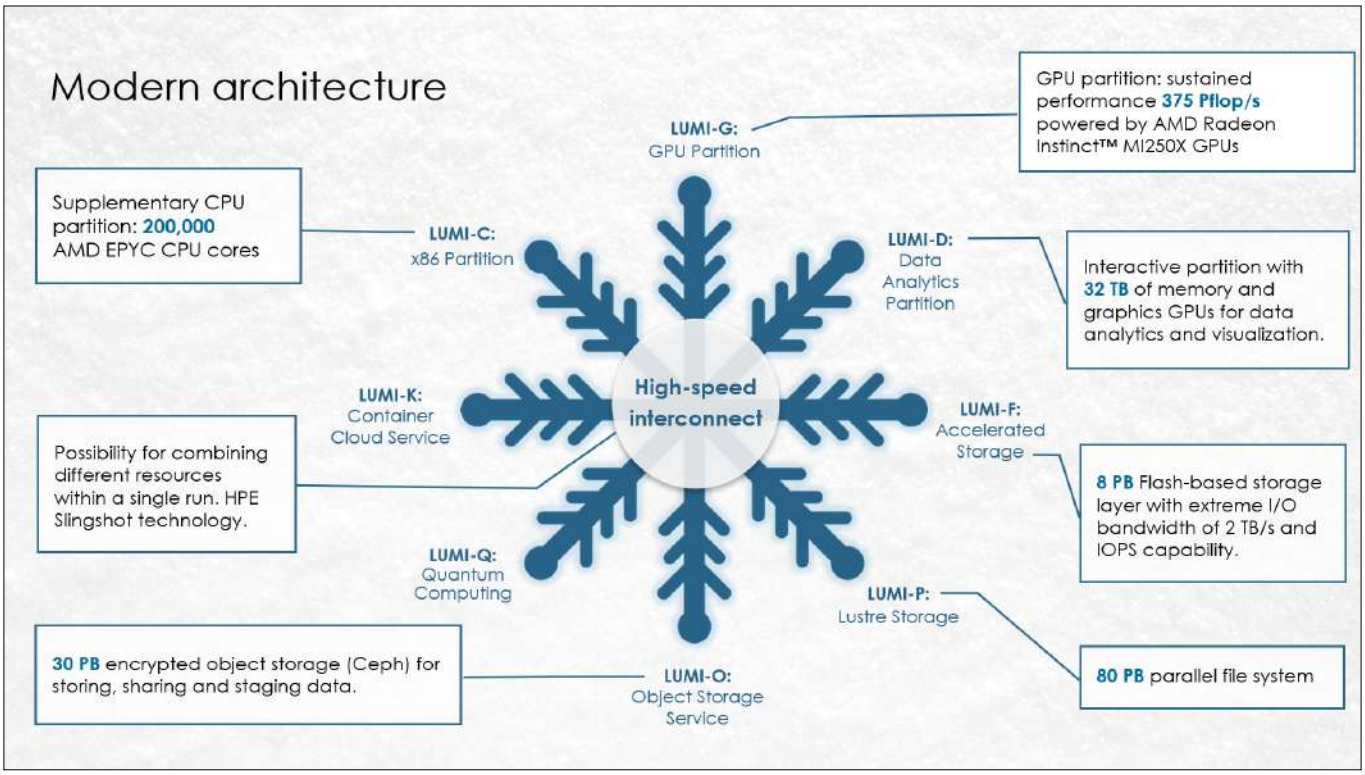


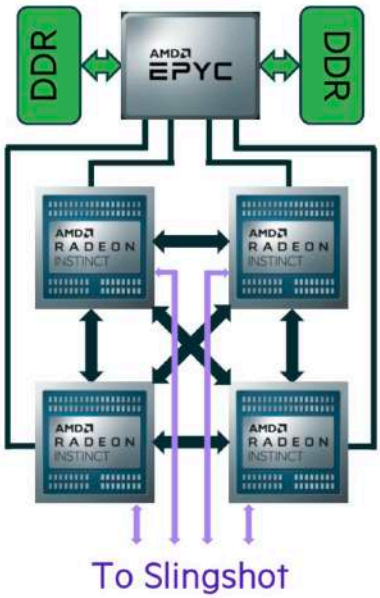


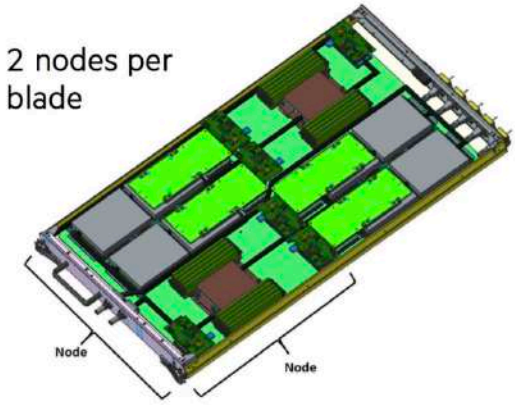
3 - Lumi - HPE Cray EX235a

- Site: EuroHPC/CSC
- Familia de sisteme: HPE Cray EX235a
- Procesor: AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X
- OS: HPE Cray OS
- Arhitectura: MPP
- Aplicatii: Cercetare
- Anul instalarii: 2023
- Numar de core-uri: 2,220,288
- Rmax(PFlops): **309**,1
- Rpeak(PFlops): **428**,7
- HPCG(PFlops): **3,**4
- Consum: **6.015** kW
- Interconectare: Slingshot-11





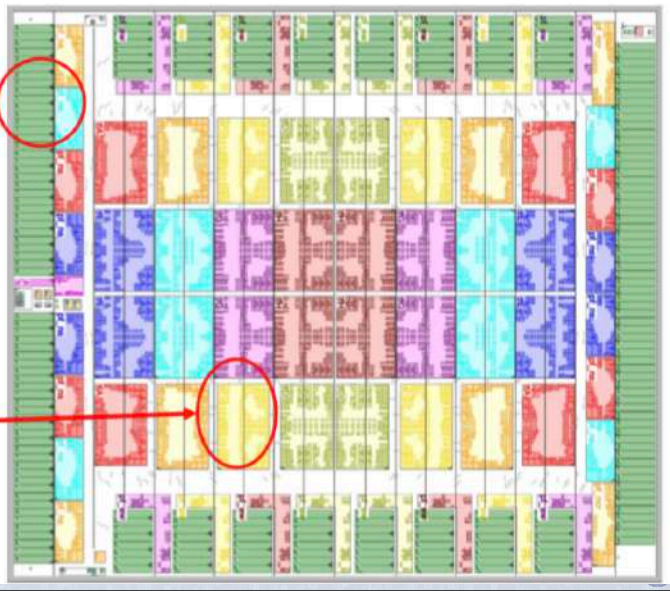




Node Architecture

Slingshot 11

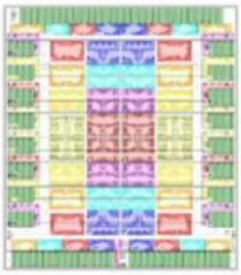
- TSMC 16nm FF
- 64 ports x 200 Gbps/dir
  - PAM4 56G
- ~250W
- Tiled architecture:
  - 32 peripheral function blocks
    - Network SerDes, MAC/PCS/LLR
    - Ethernet Lookup functions
  - 32 tile blocks
    - All other port functionality
  - Management Block (MB)
- First silicon Sept '18, Production Q4'19



Slingshot Interconnect

Rosetta Switch

- Multiple QoS levels
- Aggressive adaptive routing
- Advanced congestion control
- Very low average and tail latency
- High performance multicast and reduction



64 ports x 200 Gbps

SS-10 (100Gb)
Injection: ~14 TB/s
Bisection: ~24 TB/s

SS-11 (200Gb)
Injection: ~28 TB/s
Bisection: ~24 TB/s



Mellanox ConnectX NIC

Slingshot 10

- HPE Cray MPI stack
- Ethernet functionality
- RDMA offload



Cassini NIC

Slingshot 11

- MPI hardware tag matching
- MPI progress engine
- One-sided operations
- Collectives
- 2X injection bandwidth

Slingshot 11

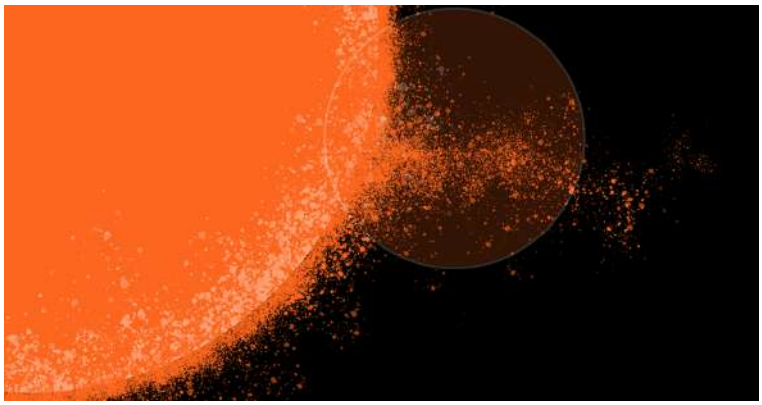




- Liquid or air cooled 1U form-factor
- Up to 8 blades / chassis & up to 64 blades / cabinet
- 100-200 Gb/s in a TOR pattern
- 1.2 B packets / s (600M / direction)

Lumi R&D Highlight

- Information from under the Sun's surface about the emergence of solar storms
  - Predict solar storms

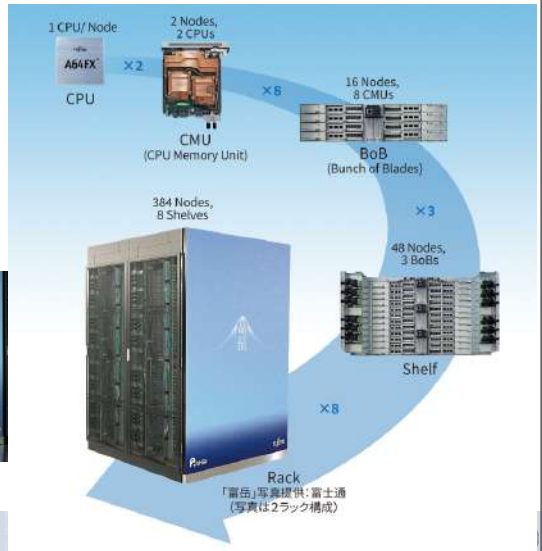


https://www.lumi-supercomputer.eu/researchers-discovered-information-from-under-the-suns-surface-about-the-emergence-of-solar-storms-using-the-lumi-supercomputer/

2 Fugaku K Computer Update

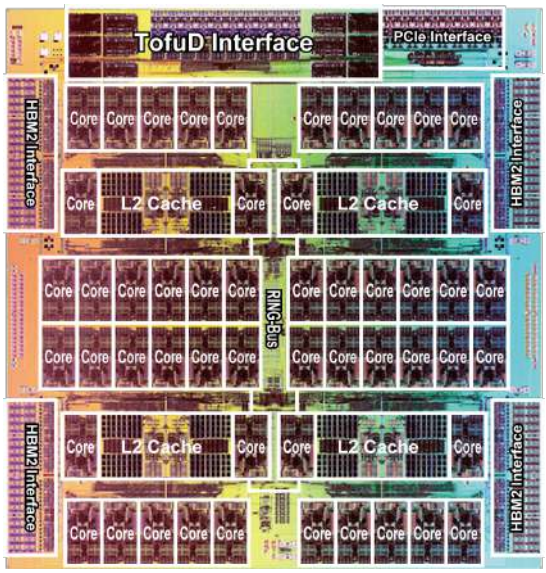
- Site: RIKEN Advanced Institute for Computational Science
- Familia de sisteme: Fujistu – SPARC64
- Procesor: A64FX 48C 2.2GHz
- OS: Linux
- Arhitectura: MPP
- Aplicatii: Cercetare
- Anul instalarii: 2020
- Numar de core-uri: 7.630.848
- Rmax(PFlops): **442**
- Rpeak(PFlops): **537**
- HPCG(PFlops): **16**
- Memorie: **5,087** TB
- Consum: **29.899** kW
- Interconectare: Custom – Tofu

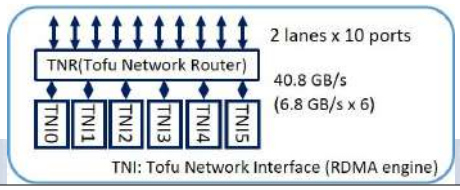




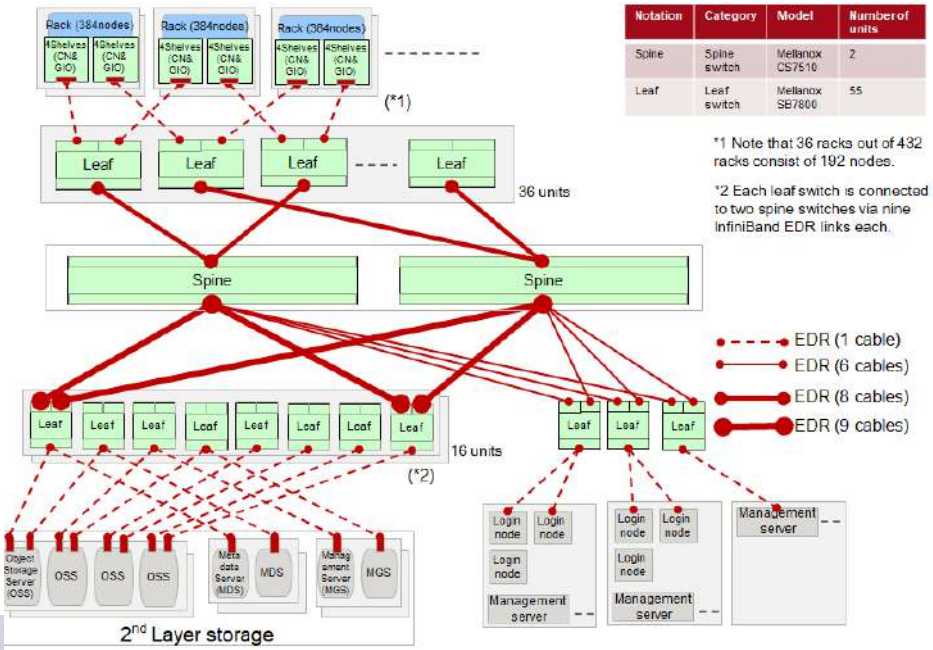
Node & Interconnect Stats

- 48+2 cores @2-2.2GHz 7nm FinFET
- Cache:
  - L1D/core: 64 KiB, 4way, 256 GB/s (load), 128 GB/s (store)
  - L2/CMG: 8 MiB, 16way
  - L2/node: 4 TB/s (load), 2 TB/s (store)
  - L2/core: 128 GB/s (load), 64 GB/s (store)
- Memory 1024 GB/s





I/O Networking



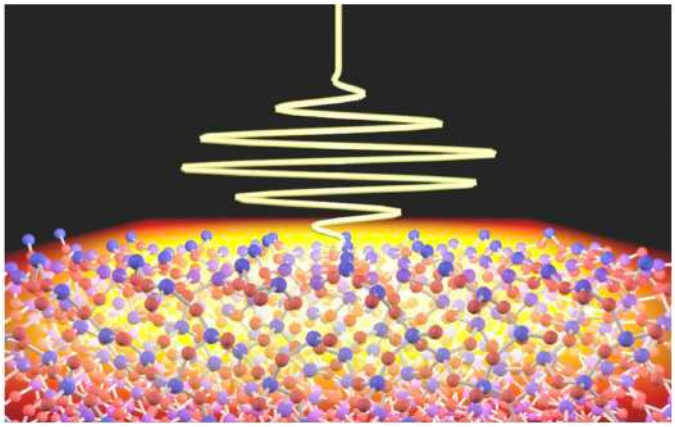
Notation	Category	Model	Number of units
Spine	Spine switch	Mellanox CS7510	2
Leaf	Leaf switch	Mellanox SB7800	55

*1 Note that 36 racks out of 432 racks consist of 192 nodes.

*2 Each leaf switch is connected to two spine switches via nine InfiniBand EDR links each.

Fugaku R&D Highlights

- COVID-19 Projects
  - Drug candidates
  - Pandemic Phenomena
- Model interactions between matter and light at the atomic scale



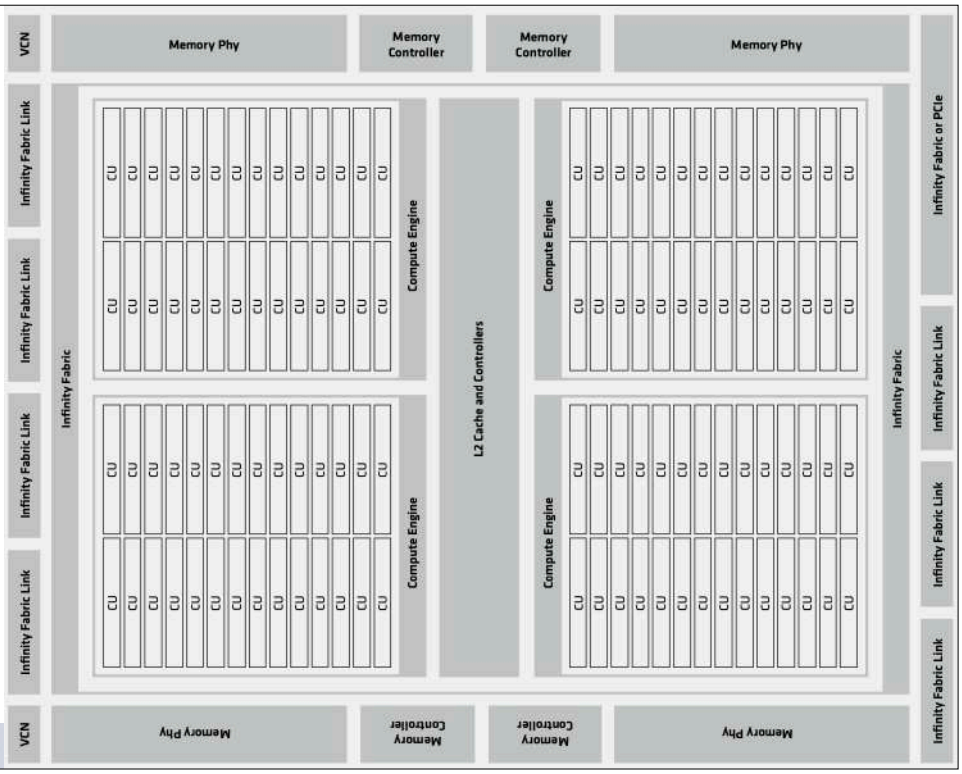
https://www.r-ccs.riken.jp/en/fugaku/research/

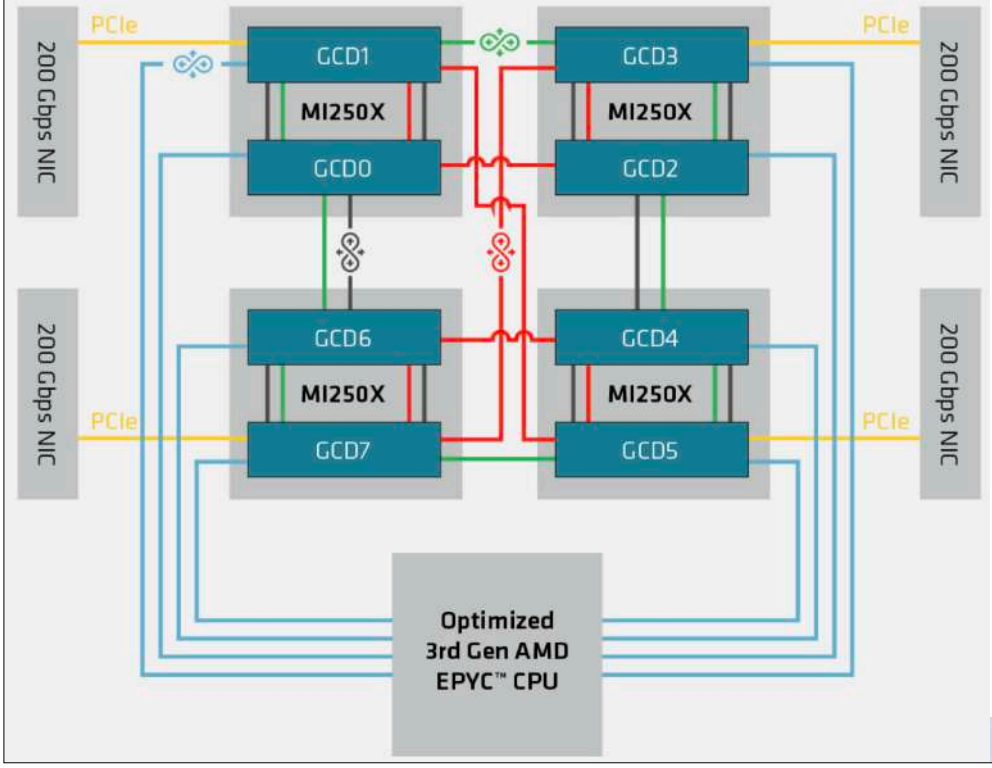
1 - Frontier - HPE Cray EX235a

- Site: DOE/SC/Oak Ridge National Laboratory
- Familia de sisteme: HPE Cray EX235a
- Procesor: AMD Optimized 3rd Gen EPYC 64C 2GHz, AMD Instinct MI250X
- OS: HPE Cray OS
- Arhitectura: MPP
- Aplicatii: Cercetare
- Anul instalarii: 2021
- Numar de core-uri: 8,699,904
- Rmax(**EFlops**): **1**,194
- Rpeak(**EFlops**): **1**,679
- HPCG(PFlops): **14,**05
- Memorie: **3,1** PB
- Consum: **22.703** kW
- Interconectare: Slingshot-11

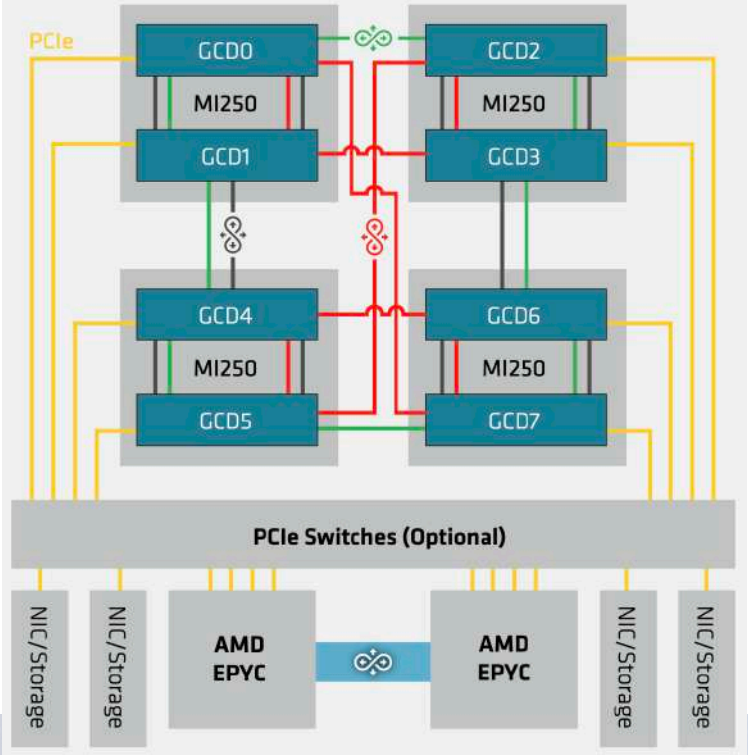


AMD Instinct MI250X





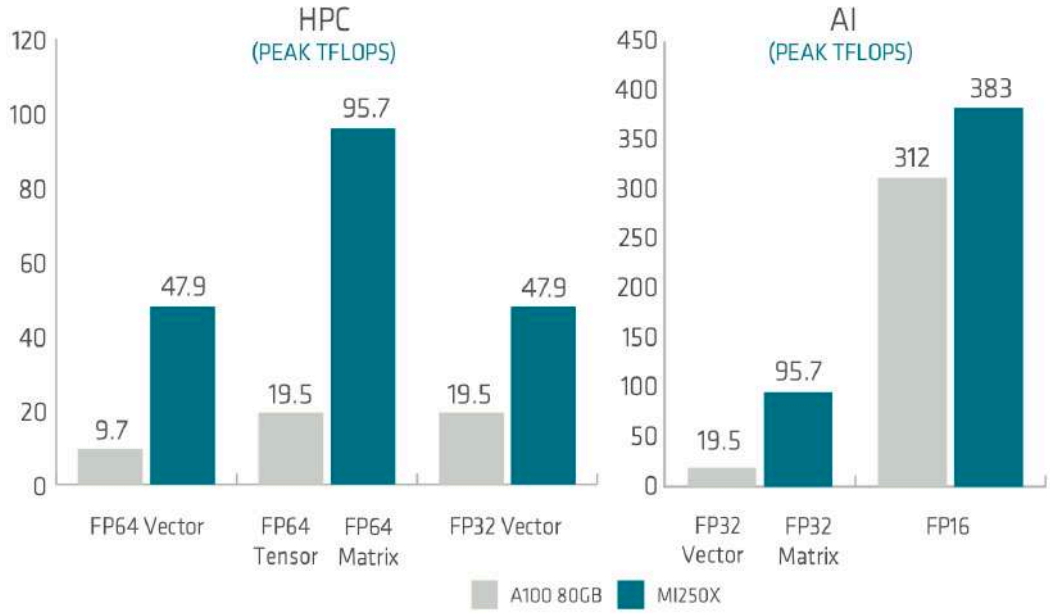
AMD Instinct MI250X HPC Node





AMD Instinct MI250X HPC/ML Node

Performance	MI210	MI250	MI250X
CDNA2 Graphics Compute Die (GCD)	1	2	2
Compute Units	104 CU	208CU	220CU
Stream processors	6,656	13,312	14,080
Matrix Cores	416	832	880
Peak FP64/FP32 Vector	22.6 TF	45.3 TF	47.9 TF
Peak FP64/FP32 Matrix	45.3 TF	90.5 TF	95.7 TF
Peak FP16/BF16	181.0 TF	362.1 TF	383.0 TF
Peak INT4/INT8	181.0 TOPS	362.1 TOPS	383.0 TOPS
Memory			
Memory Size	64GB HBM2e	128GB HBM2e	128GB HBM2e
Memory Interface	4,096 bits	8,192 bits	8,192 bits
Memory Clock	1.6GHz	1.6GHz	1.6GHz
Memory Bandwidth	up to 1.6 TB/sec	up to 3.2TB/sec2	up to 3.2TB/sec2
Scalability			
Infinity Fabric Links	up to 3	up to 6	up to 8
xGMI Bridge Card Configuration	Yes (Dual | Quad Hives)	NA	NA
Coherency Enabled	No	No	Yes



Graph 1: Peak TFLOPS across range of mixed-precision Compute[1]



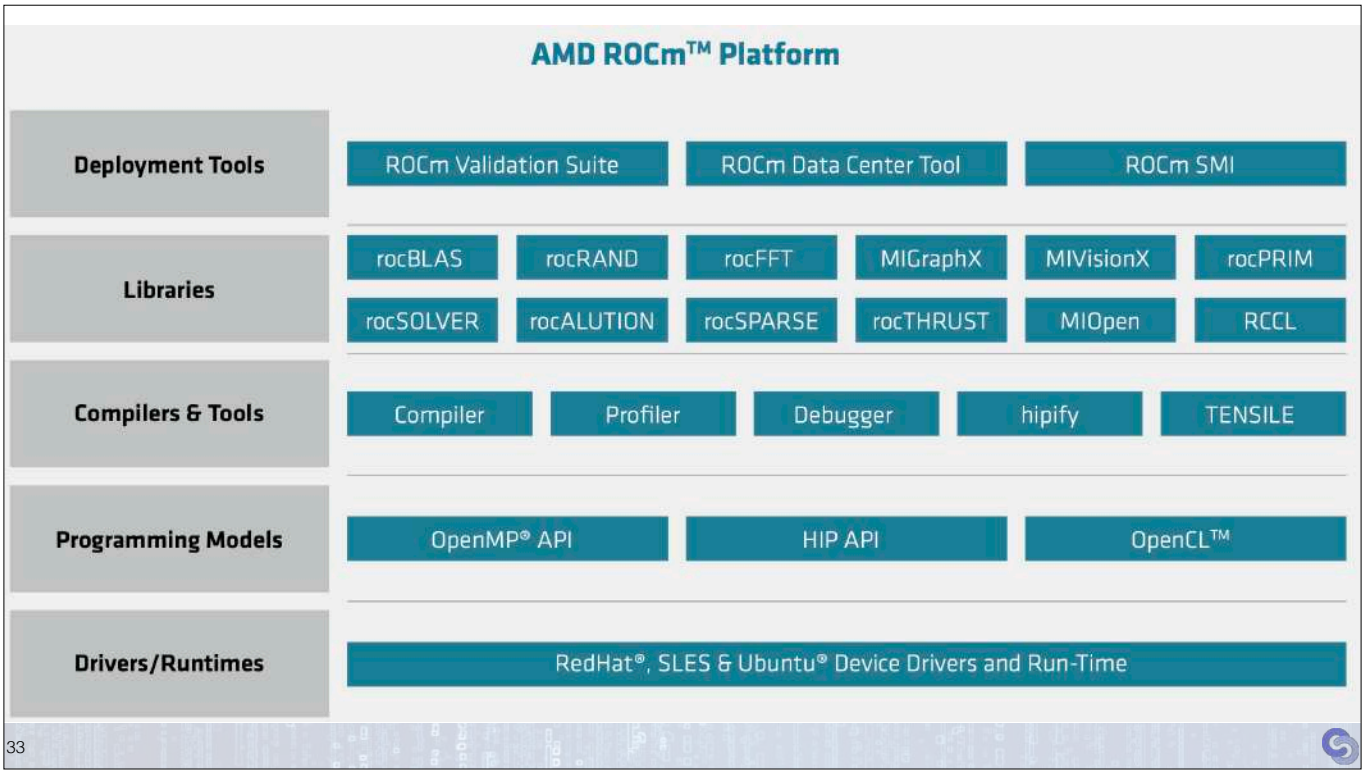
Original

```
float vxi = 0.0f, vyi = 0.0f, vzi = 0.0f;
for (int j = threadIdx.x; j < count1; j += blockDim.x) {
  float dx = xx1[j] - xxi;
  float dy = yy1[j] - yyi;
  float dz = zz1[j] - zzi;
  float dist2 = dx*dx + dy*dy + dz*dz;
  if (dist2 < fsrrmax2) {
    float rtemp = (dist2 + rsm2)*(dist2 + rsm2)*(dist2 + rsm2);
    float f_over_r = massi*mass1[j]*(1.0f/sqrt(rtemp) -
        (ma0 + dist2*(ma1 + dist2*(ma2 + dist2*(ma3 +
        dist2*(ma4 + dist2*ma5))))));
    vxi += fcoeff*f_over_r*dx;
    vyi += fcoeff*f_over_r*dy;
    vzi += fcoeff*f_over_r*dz;
  }
}
```

Modified to use Packed FMA32

```
float vxi = 0.0f, vyi = 0.0f, vzi = 0.0f;
  for (int j = threadIdx.x; j < count1; j += 2*blockDim.x) {
    float2 dx = {xx1[j] - xxi, xx1[j + blockDim.x] - xxi};
    float2 dy = {yy1[j] - yyi, yy1[j + blockDim.x] - yyi};
    float2 dz = {zz1[j] - zzi, zz1[j + blockDim.x] - zzi};
    float2 dist2 = dx*dx + dy*dy + dz*dz;
    bool check[2] = {dist2.x < fsrrmax2, dist2.y < fsrrmax2};
    if (check[0] || check[1]) {
      float2 rtemp = (dist2 + rsm2)*(dist2 + rsm2)*(dist2 + rsm2);
      float2 mass1_2 = {mass1[j],mass1[j + blockDim.x]};
      float2 sqrt_rtemp = {sqrtf(rtemp.x),sqrtf(rtemp.y)};
      float2 f_over_r = massi*mass1_2*(1.0f/sqrt_rtemp -
          (ma0 + dist2*(ma1 + dist2*(ma2 + dist2*(ma3 +
          dist2*(ma4 + dist2*ma5))))));
      float2 vxi_tmp = fcoeff*f_over_r*dx;
      float2 vyi_tmp = fcoeff*f_over_r*dy;
      float2 vzi_tmp = fcoeff*f_over_r*dz;
      vxi += check[0] ? vxi_tmp.x : 0.0f;
      vxi += check[1] ? vxi_tmp.y : 0.0f;
      vyi += check[0] ? vyi_tmp.x : 0.0f;
      vyi += check[1] ? vyi_tmp.y : 0.0f;
      vzi += check[0] ? vzi_tmp.x : 0.0f;
      vzi += check[1] ? vzi_tmp.y : 0.0f;
    }
  }
```

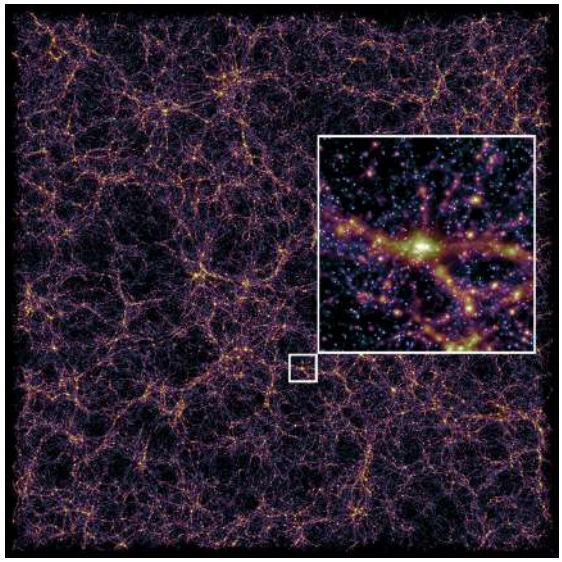


Frontier R&D Highlight

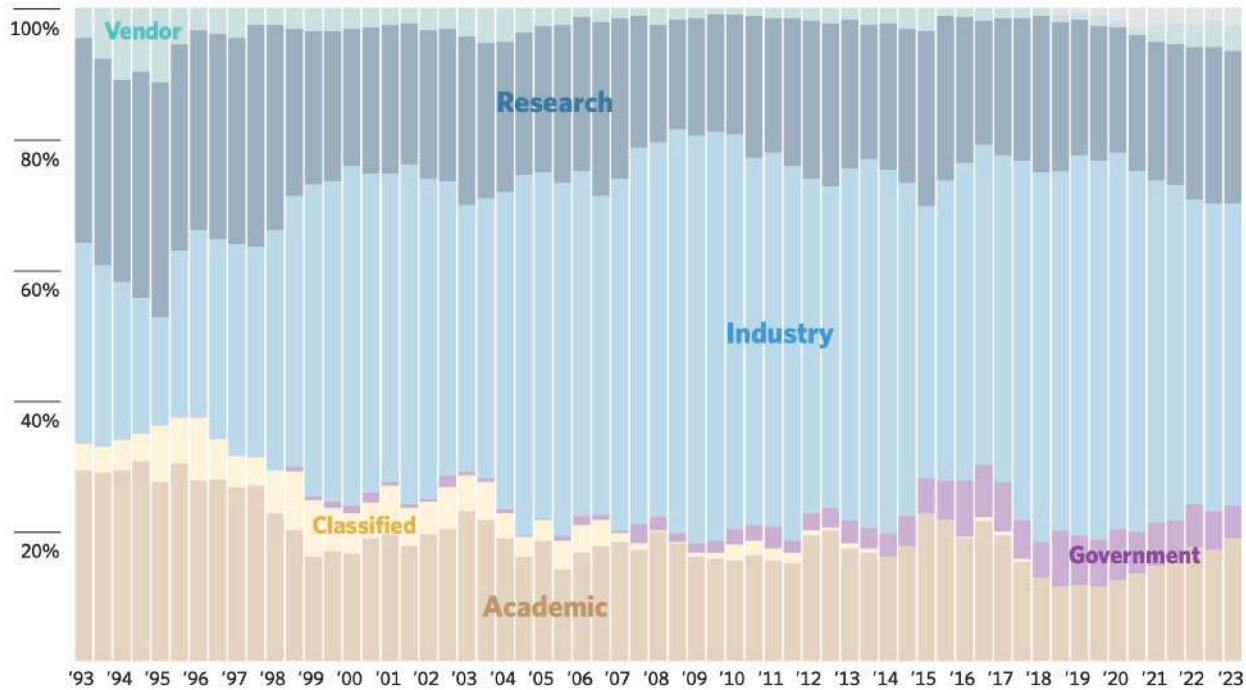
- More detailed Universe
- HACC (Hardware/Hybrid Accelerated Cosmology Code)
- Port from NVidia GP-GPUs to AMD Instinct

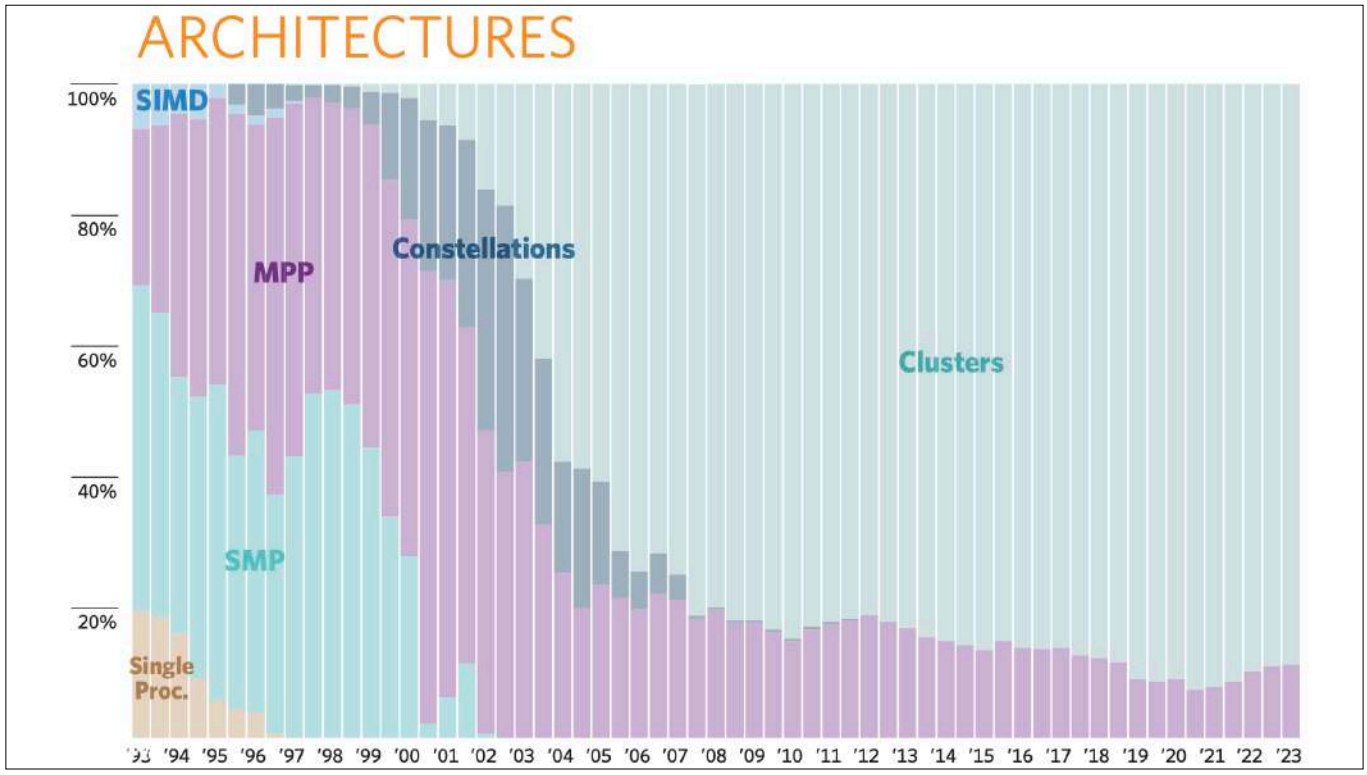
https://www.olcf.ornl.gov/2023/04/26/simulating-a-more-detailed-universe-with-frontier/

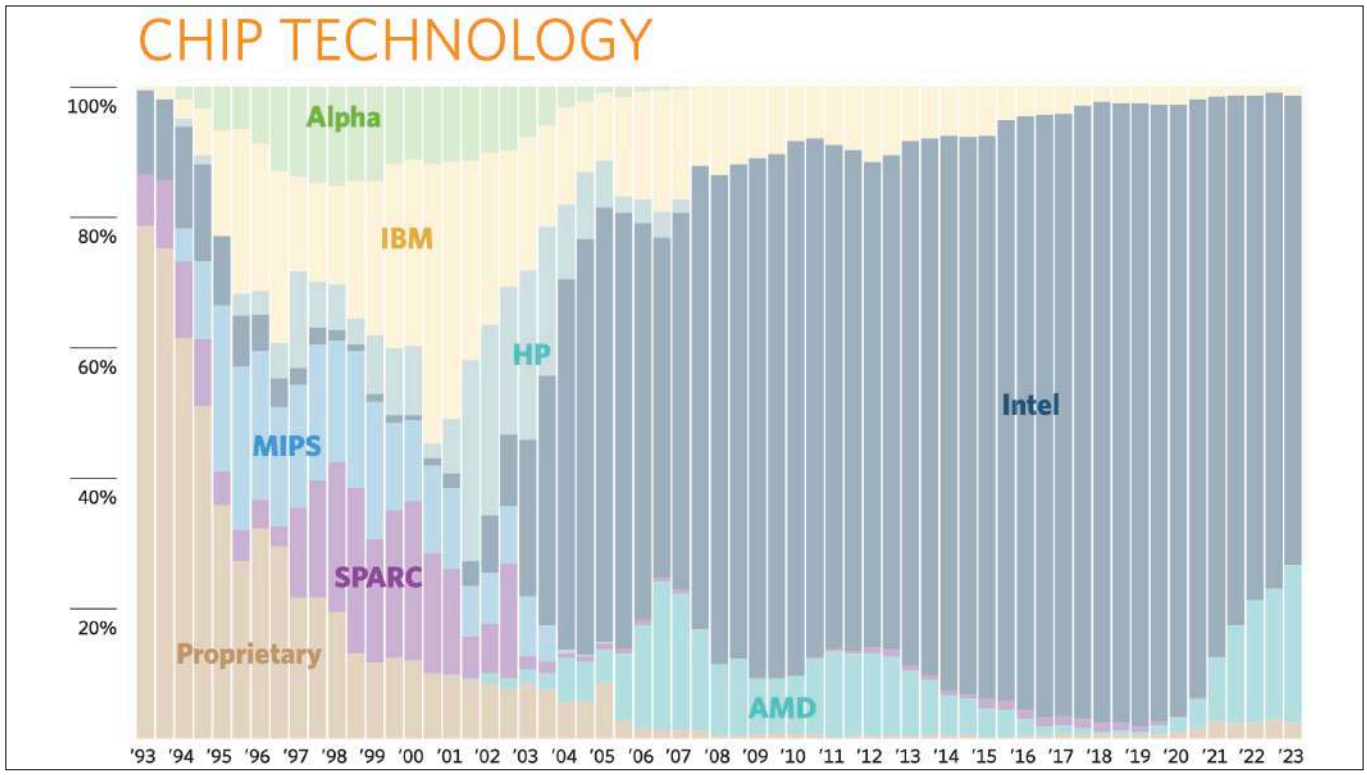




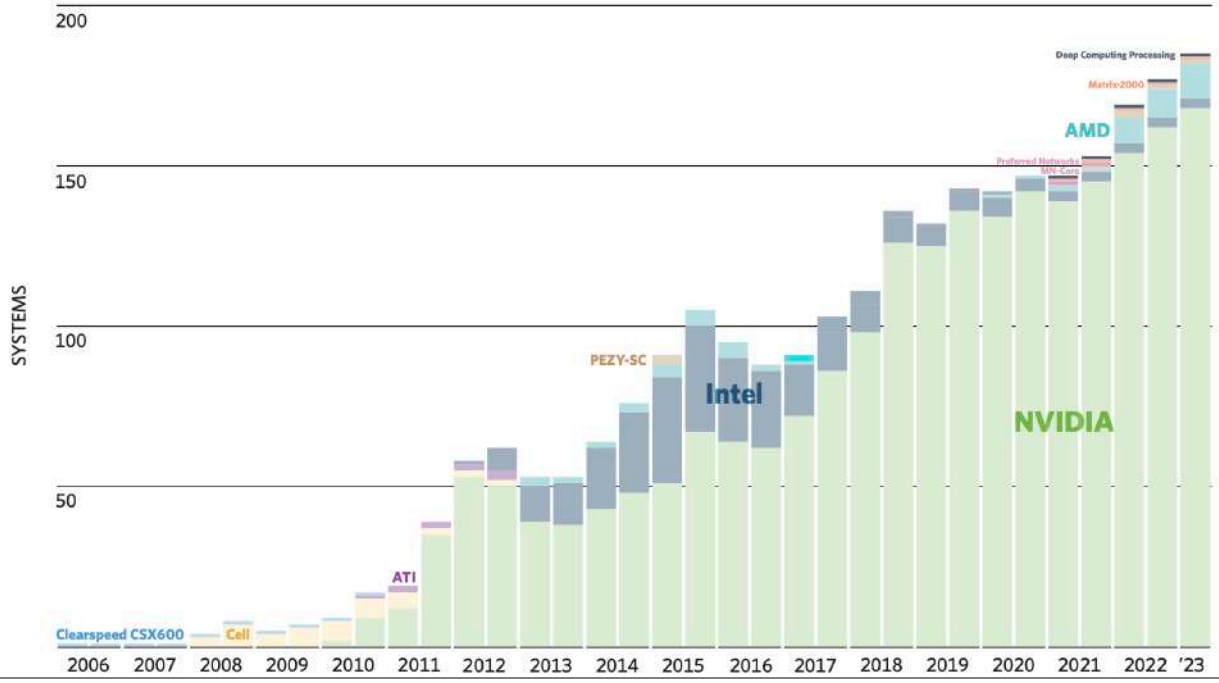
Concluzii Top 500




ARCHITECTURES
100%
80%
60%
40%
20%
SIMD
MPP
Constellations
Clusters
SMP
Single Proc.
'93 '94 '95 '96 '97 '98 '99 '00 '01 '02 '03 '04 '05 '06 '07 '08 '09 '10 '11 '12 '13 '14 '15 '16 '17 '18 '19 '20 '21 '22 '23


CHIP TECHNOLOGY
100%
80%
60%
40%
20%
Alpha
IBM
HP
Intel
MIPS
SPARC
Proprietary
AMD
'93 '94 '95 '96 '97 '98 '99 '00 '01 '02 '03 '04 '05 '06 '07 '08 '09 '10 '11 '12 '13 '14 '15 '16 '17 '18 '19 '20 '21 '22 '23

ACCELERATORS/CO-PROCESSORS



Vendors

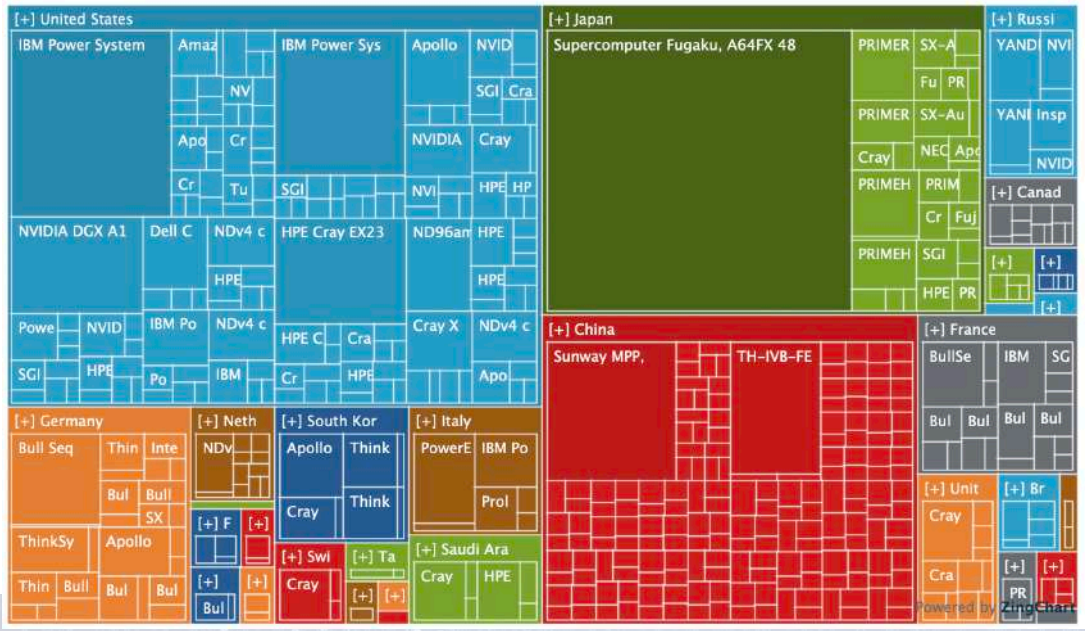
		Count	System Share (%)	Rmax (TFlops)	Rpeak (TFlops)	Cores
1	Lenovo	173	34.6	330,547	610,259	12,339,312
2	Sugon	71	14.2	119,286	301,610	5,182,664
3	Inspur	66	13.2	102,121	219,012	2,666,744
4	HPE	35	7	122,353	179,809	3,174,784
5	Cray/HPE	35	7	162,476	249,470	6,029,800
6	Atos	23	4.6	64,828	103,337	2,117,592
7	Fujitsu	13	2.6	60,538	96,398	1,589,640
8	IBM	13	2.6	216,919	287,902	6,005,272
9	Penguin Computing	11	2.2	20,034	25,248	614,136
10	Dell EMC	11	2.2	52,309	89,404	1,276,988

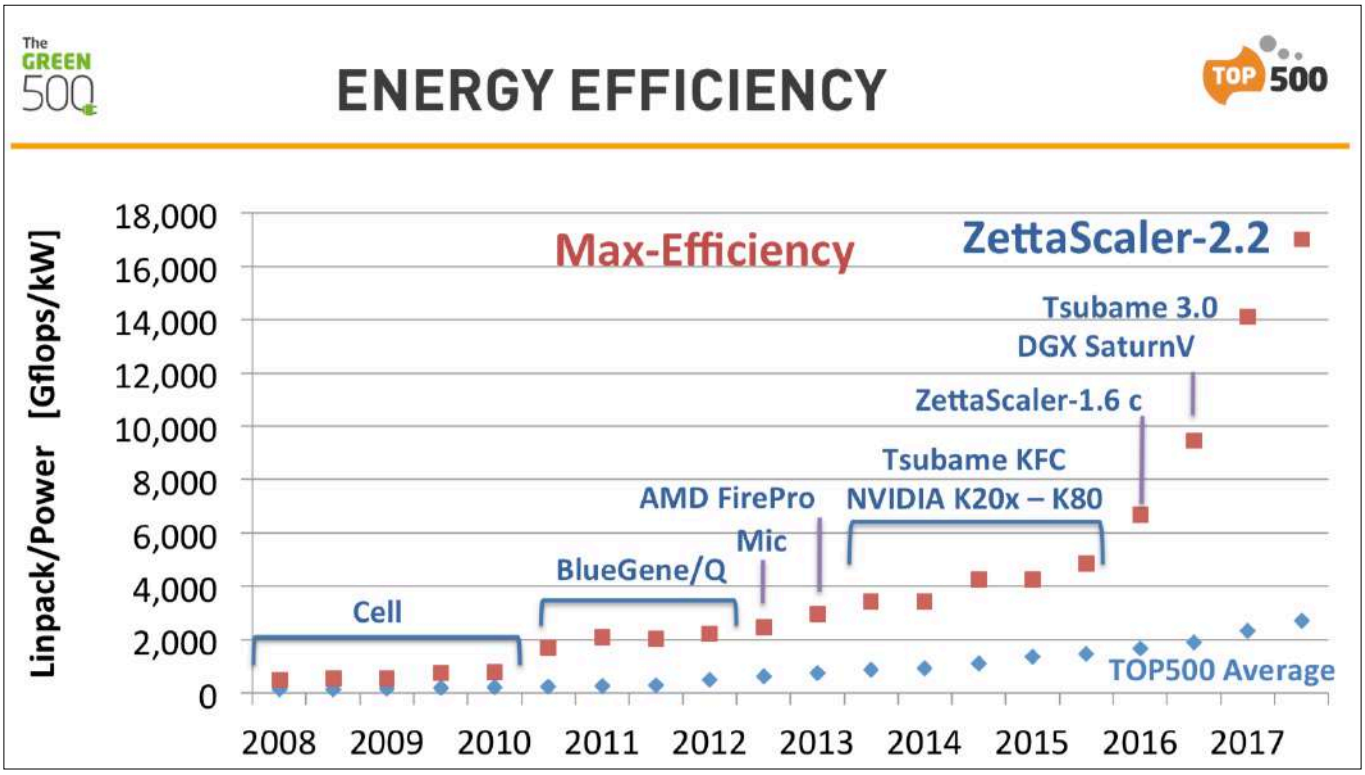


Interconnect

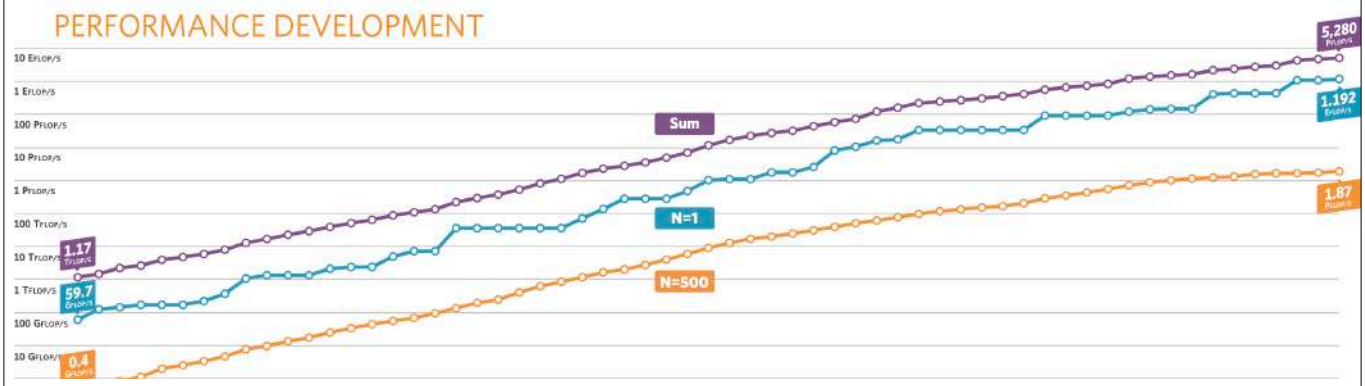
		Count	System Share (%)	Rmax (TFlops)	Rpeak (TFlops)	Cores
1	Gigabit Ethernet	259	51.8	430,575	897,359	16,913,772
2	Infiniband	140	28	662,896	1,013,372	15,946,634
3	Omnipath	50	10	181,500	285,493	4,704,264
4	Custom Interconnect	45	9	356,933	524,811	24,991,220
5	Proprietary Network	5	1	13,009	16,406	508,496
6	Myrinet	1	0.2	1,975	6,595	89,600

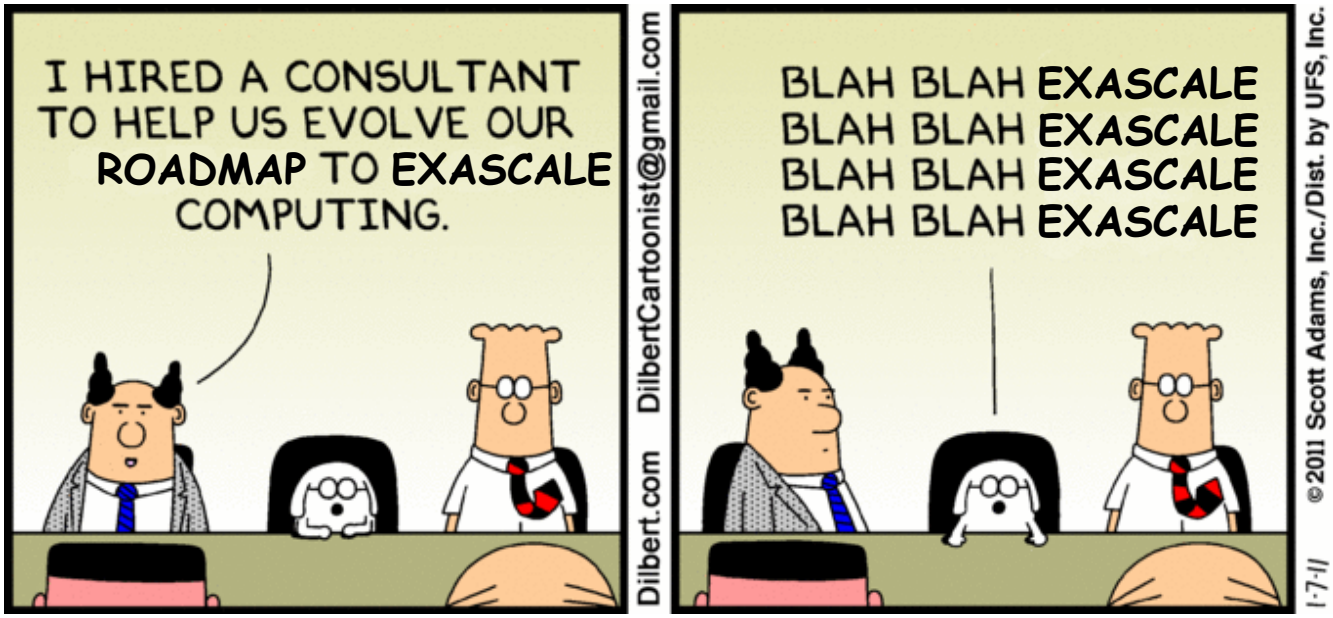
Instalari @Tari





Performanta in timp…







Feedback @ASC

Ce am invatat in aces curs?

- Top 5
- Concluzii Top 500



Thank you for your attention.





emil.slusanschi@cs.pub.ro
cs.pub.ro